Cube-Sum Permutations: Exact Membership and Computation-Free Exponential Families

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Abstract

Let $P(n)$ count permutations p of $\{1,\dots,n\}$ such that $i+p_i$ is a perfect cube for every index i. We give a computer-assisted resolution of the existence conjecture recorded as OEIS A006063, as well as the exact membership classification for A096680: $P(n)>0$ for every $n\geq 173$, while $P(173)=P(174)=1$ and $P(n)\geq 3$ for every $n\geq 175$. An explicit seven-block interval involution at $n=173$ and only 1844 one-digit cube roots in 191 alternating-path words certify every remaining base size up to 364. An elementary reflection induction covers all larger sizes; the earlier 109-permutation and 60-seed certificates are retained only for independent cross-checking. Exact Hall-set certificates classify the earlier failures, and alternating-cycle certificates establish multiplicity. For every fixed $s\geq 2$ we prove an s-branch reflection-entropy inequality, giving the stronger limit

$$\liminf_{n\to\infty}\frac{\log P(n)}{(2n)^{1/3}}\geq\frac{\log 2}{2}.$$

More strongly, we identify a two-reflection, three-cycle identity gadget that is entirely independent of these finite certificates. An explicit weighted identity of four cubes yields

$$P(681707u^3)\geq 2^{148877u^3-1}\qquad(u\geq 1),$$

and hence

$$\limsup_{n\to\infty}\frac{\log P(n)}{n}\geq\frac{148877\log 2}{681707}.$$

These permutations also contain exact binary Hamming codes of linear dimension and distance. A conditional analogue of the tensorization theorem holds for all perfect-power exponents. All finite verification data and exact checkers are supplied separately; literature priority and external independent review remain open.

1 The reflection principle

Write $\mathcal{C}=\{k^3:k=2,3,\dots\}$ and let $P(n)$ be the number of cube-sum permutations of $\{1,\dots,n\}$. By convention the empty permutation is permitted only in the general extension lemma; our classification concerns $n\geq 1$. Let $\mathcal{G}=\{n\geq 1:P(n)>0\}$.

Lemma 1 (Reflection and counting)**.** *If $0\leq m<n$ and $c=m+n+1$ is a cube, every valid permutation p on $\{1,\dots,m\}$ extends to a valid permutation q on $\{1,\dots,n\}$ via*

$$q_i=\begin{cases}p_i, & i\leq m,\\ c-i, & m<i\leq n.\end{cases}$$

For different admissible cubes c, the images of these extensions are disjoint. In particular,

$$P(n)\geq\sum_{\substack{c\in\mathcal{C}\\ n+1\leq c\leq 2n}}P(c-n-1),\tag{1}$$

with the convention $P(0)=1$.

Proof. Because $c = m+n+1$, reflection $i \mapsto c-i$ permutes precisely the new interval $\{m+1, \dots, n\}$, and $i+q_i = c$ there. Old indices retain their cube sums. The embedding is injective for each c and two different c yield different last entries $q_n = c - n$. The claimed sum counts disjoint families. □

Lemma 2 (The cube-gap bound)**.** *For each $n \geq 365$ some cube c satisfies $n + 174 \leq c \leq 2n$. For each $n \geq 556$ some cube c satisfies $n + 176 \leq c \leq 2n$.*

Proof. For $365 \leq n \leq 555$, take $c = 729$. For $n \geq 556$, let k be the least integer with $k^3 \geq n + 176$; then $k \geq 10$ and $n \geq (k-1)^3 - 175$. Thus

$$2n - k^3 \geq k^3 - 6k^2 + 6k - 352 \geq 0.$$

The final polynomial equals 108 at $k = 10$ and has strictly positive successive differences $3k^2 - 9k + 1$ for $k \geq 10$. Hence the second assertion holds and implies the first for $n \geq 556$. □

2 A short, local existence certificate

The preceding reflection argument is infinite and elementary. The finite existence verification can itself be made much more transparent than an array of 192 long permutations: we give *one* explicit involution at size 173, then 191 short alternating-path words. Only 1844 single-digit cube roots are needed.

Lemma 3 (An explicit order-173 involution)**.** *There is a cube-sum permutation on $[1, 173]$, given on each listed interval I by $p(i) = c - i$. Intervals paired by a double arrow use the same cube c:*

intervals I	c
$[1, 7]$	8
$[22, 42]$	64
$[106, 110]$	216
$[8, 12] \longleftrightarrow [15, 19]$	27
$[13, 14] \longleftrightarrow [111, 112]$	125
$[20, 21] \longleftrightarrow [104, 105]$	125
$[43, 103] \longleftrightarrow [113, 173]$	216

Proof. The eleven intervals partition $[1, 173]$. Reflection $i \mapsto c - i$ preserves each of the first three intervals and exchanges the two intervals in each of the last four rows. Thus it defines an involution, and every sum $i + p(i)$ is a cube. □

Lemma 4 (An alternating-word extension)**.** *Suppose p is a cube-sum permutation of $[1, n-1]$. Let $w = r_1 \cdots r_t$ be a finite word of positive integer cube roots. Starting from $u_0 = n$, define $v_j = r_j^3 - u_{j-1}$. For $j < t$, require $v_j \in [1, n-1]$ and let $u_j = p^{-1}(v_j)$. Suppose all these $u_0, u_1, \dots, u_{t-1}$ are distinct, all $v_1, \dots, v_t$ are distinct, and $v_t = n$. Then replacing the old matching edges by*

$$q(u_{j-1}) = v_j \quad (1 \leq j \leq t),$$

while keeping all other old assignments, yields a cube-sum permutation of $[1, n]$.

Proof. The old matching edges removed along this path are $u_j \mapsto v_j$ for $j < t$; these form a bijection between the old visited left and right vertices. Their replacements join u_{j-1} to v_j and have sum r_j^3 by construction. The new left vertex n and the new right vertex n are incorporated, without repeating any endpoint. Every untouched vertex retains its old matching edge. □

The `augmenting_cube_words.txt` appendix lists one word for each $n = 174, \dots, 364$. Every root is a single decimal digit in $\{2, \dots, 8\}$. The words are checked using Lemma 4 and the predecessor matching obtained from the previous word; this is *local* verification, not a matching search or a scan to a large cutoff. The independent, standard-library-only `verify_word_paths.py` checks all required edge equalities and endpoint distinctness. The seven-line interval seed plus 1844 root digits certifies all 192 base cases of Theorem 1; combined with the cube-gap induction this proves existence for *every* $n \geq 173$. The older long witness archives remain as secondary independent cross-checks, not as prerequisites of this compact proof.

3 Finite certificates and exact membership

Our finite data are provided as UTF-8 JSON arrays of integers. A valid witness p is checked solely by

$$\mathrm{sort}(p) = (1, \ldots, n), \qquad \forall i \quad i + p_i \in \mathcal{C}.$$

For a negative instance, let $N_n(S) = \{j \in [1, n] : \exists i \in S,\ i + j \in \mathcal{C}\}$. Hall's marriage theorem gives the immediate obstruction

$$|N_n(S)| < |S| \quad \Longrightarrow \quad P(n) = 0. \tag{2}$$

The file `exact_classification_below_173.json` provides 48 explicit valid permutations and 124 explicit Hall-deficient subsets, exactly partitioning $1 \le n \le 172$. Every assertion is directly checked by `verify_extended.py`. For instance, for $n = 172$ the 27 left indices

$$S = \{1, 5, 8, 13, 20, 31, 35, 38, 42, 43, 45, 92, 96, 99,$$
$$104, 111, 122, 126, 129, 133, 134, 136, 153, 157, 160, 165, 172\}$$

have precisely 26 distinct possible right neighbors. Thus $172 \notin \mathcal{G}$, making the threshold below sharp.

For the unbounded tail, only the 109 witness sizes

$$173 \le n \le 255, \qquad 339 \le n \le 364 \tag{3}$$

require explicit stored permutations. Indeed for each $256 \le n \le 338$, put $m = 511 - n \in [173, 255]$; then $m + n + 1 = 512 = 8^3$, so Lemma 1 constructs the missing 83 cases. This compresses the previously saved 192 base cases by $83/192 > 43\%$. All 109 stored witnesses in `core_109.json`, and all 192 reconstructed witnesses, are checked with exact integer arithmetic. The earlier full 192-row file is retained as an additional independent-format audit source.

There is a further useful compression when the 48 verified positive sizes below 173 are reused. An ascending reflection-closure algorithm starts from those small solutions and, at each $173 \le n \le 364$, uses any already certified $m < n$ for which $m + n + 1$ is a cube. Precisely 132 lengths are obtained by reflection and only 60 further lengths need explicit witnesses. Here the empty matching is allowed: $n = 342$ follows directly from $c = 343$, with $p(i) = 343 - i$. The 60 additional witnesses are archived in `core_60_with_empty.json`, and `verify_extended.py` certifies this alternate compression independently of the 109-case presentation.

Proposition 1 (Limit of reflection-only compression)**.** *Suppose the empty matching and the* 48 *known positive lengths below* 173 *are provided as initial data. Among proofs of all lengths* $173, \ldots, 364$ *that use only the reflection extension of Lemma 1 and independently supplied new full-length seed permutations, at least* 60 *extra seeds are necessary, and* 60 *suffice.*

Proof. Process $n = 173, \ldots, 364$ in ascending order. If no cube c yields a previously certified positive predecessor $m = c - n - 1 \in [0, n - 1]$, then reflection cannot establish n from earlier data, so a new seed is forced. Otherwise choose such a predecessor and perform reflection, without adding a seed. This deterministic closure rule yields 44 forced seeds in $[173, 255]$, none in $[256, 338]$ (use $c = 512$ and $m = 511 - n$), and 16 in $[339, 364]$. Thus $44 + 16 = 60$, and the attached short integer-only checker reproduces the forced index list. Every forced seed is unavoidable by the definition of the restricted proof rule. □

This is *not* a lower bound on arbitrary mathematical proofs. The alternating-path extension in Lemma 4 is a different operation and indeed replaces those 60 large independent seeds by one explicit involution and a small word table.

Theorem 1 (Complete A006063 membership)**.** *For every integer* $n \ge 173$*,* $P(n) > 0$*. Below* 173*, the complete membership list is*

$$\mathcal{G} \cap [1, 172] = \{7, 19, 26, 37, 44, 56, 63, 66, 68, 80, 82, 85, 87, 98, 100,$$
$$103, 105, 110, 112, 115, 116, 117, 118, 119, 120, 121,$$
$$122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132,$$
$$133, 135, 147, 149, 150, 151, 152, 155, 156, 159, 171\}.$$

In sequence-index notation this gives

$$\boxed{\mathrm{A006063}(j) = j + 124 \quad (j \ge 49).}$$

Proof. The exact finite witnesses establish $173 \le n \le 364$, and the Hall certificates settle the smaller sizes. For $n \ge 365$, Lemma 2 gives a cube $c \in [n+174, 2n]$. The predecessor $m = c - n - 1$ satisfies $173 \le m < n$. By strong induction $P(m) > 0$, and Lemma 1 yields $P(n) > 0$. There are precisely 48 positive indices below 173, so the 49th and all later sequence terms are consecutive. □

4 Multiplicity and the second OEIS classification

A cube-sum permutation p is a perfect matching in the bipartite graph with left and right vertices $1, \dots, n$, where (i, j) is an edge exactly when $i + j \in \mathcal{C}$. For any matching p, form a directed graph D_p on the left vertices, putting $i \to j$ whenever $i \ne j$ and (i, p_j) is a cube-sum edge.

Lemma 5 (Alternating-cycle criterion)**.** *For a valid permutation p, there exists another distinct valid permutation if and only if D_p has a directed cycle.*

Proof. Along a directed cycle $i_1 \to \cdots \to i_t \to i_1$, replace p_{i_s} by $p_{i_{s+1}}$ (cyclic indices). Every moved pair is an allowed edge, and all right vertices remain distinct. Conversely, if $q \ne p$ is another valid permutation, the nontrivial cycles of the permutation $p^{-1} \circ q$ consist of directed cycles in D_p. Thus acyclicity is a rigorous uniqueness certificate. □

The file `second_solutions.json` records directed cycles, not complete duplicate permutations. Direct finite checking establishes: among the 48 positive lengths below 173, exactly

$$E = \{112, 115, 116, 117, 119, 124, 125, 126, 127, 128, 129, 130, 133\}$$

have a directed alternating cycle; every other positive length below 173, as well as 173 and 174, has an acyclic D_p and therefore a unique permutation.

Furthermore, all 190 base cases $175 \le n \le 364$ have *two distinct* alternating-cycle certificates, archived as `second_solutions.json` and `third_solutions.json`. They yield three different permutations, counting the original matching. To extend the multiplicity assertion: for $365 \le n \le 553$, the cube 729 gives $m = 728 - n \in [175, 363]$. The two exceptional lengths $554, 555$ have separate pairs of alternating-cycle certificates obtained by extending the unique solutions at $174, 173$ respectively. For $n \ge 556$, the stronger part of Lemma 2 gives $m = c - n - 1 \ge 175$. Three distinct solutions for m remain distinct after reflection, finishing an induction.

Theorem 2 (Complete A096680 membership)**.** *The positive integers with at least two cube-sum permutations are exactly*

$$\boxed{E \ \cup \ \{175, 176, 177, \dots\}.}$$

Moreover $P(n) \ge 3$ for every $n \ge 175$, while $P(173) = P(174) = 1$. The sequence terms satisfy

$$\boxed{\text{A096680}(j) = j + 161 \quad (j \ge 14).}$$

Sharpness of the constant three. The bound $P(n) \ge 3$ for all $n \ge 175$ cannot be improved to four: $P(175) = 3$ exactly. For the stored matching at 175, the directed alternation graph has precisely one nontrivial strongly connected component,

$$\{4, 11, 16, 23, 41, 48, 102, 114, 168, 175\},$$

and the other 165 vertices are singleton components. Every cycle of a permutation alternative lies in a strongly connected component, so the permanent reduces to the 10-by-10 component's allowed-edge matrix (including the matching self-edges). A direct bitmask dynamic program gives its permanent as 3. This computation is independently reproduced in `verify_extended.py` using exact integers.

5 A nine-gadget exceptional case at $n = 236$

An *involution* is a permutation satisfying $p^2 = \mathrm{id}$. At $n = 236$ there are exactly 512 cube-sum permutations, *none* of which is an involution. This follows from a short alternating-cycle decomposition, not from a general permanent computation.

Define a reference permutation p of $[1, 236]$ by $p(i) = c - i$ on the following fifteen intervals:

I	c	I	c
$[1, 7]$	8	$[8, 9]$	216
$[10, 18]$	64	$[19, 43]$	125
$[44, 45]$	216	$[46, 70]$	125
$[71, 81]$	216	$[82, 106]$	125
$[107, 134]$	343	$[135, 136]$	216
$[137, 170]$	343	$[171, 172]$	216
$[173, 197]$	343	$[198, 208]$	216
$[209, 236]$	343		

These interval images partition $[1, 236]$, and every prescribed sum is a cube.

Lemma 6 (Translation of alternating cycles)**.** *Suppose $p(i) = c_j - i$ on pairwise disjoint integer intervals I_j for $1 \le j \le r$. If $i_j \in I_j$ form a directed alternating cycle $i_1 \to i_2 \to \cdots \to i_r \to i_1$ for p, then every integer shift t with $i_j + t \in I_j$ for all j produces another alternating cycle on $i_j + t$. If t ranges through a set T of such shifts, the translated cycles are disjoint and yield at least $2^{|T|}$ distinct valid cube-sum permutations.*

Proof. Because $p(i_{j+1} + t) = c_{j+1} - i_{j+1} - t$,

$$(i_j + t) + p(i_{j+1} + t) = i_j + c_{j+1} - i_{j+1} = i_j + p(i_{j+1}),$$

so every translated alternating edge has the same cube sum as its original. Since the I_j are pairwise disjoint, different shifts cannot share a vertex within any I_j. The cycles are therefore vertex-disjoint and can be switched independently. □

In the present 236-element template,

$$10 \to 137 \to 46 \to 198 \to 71 \to 10$$

is one alternating 5-cycle, with successive cube sums

$$216, \quad 216, \quad 64, \quad 343, \quad 125.$$

The five vertices belong respectively to the five different reflection blocks $[10, 18]$, $[137, 170]$, $[46, 70]$, $[198, 208]$, $[71, 81]$. The common shifts $t = 0, \ldots, 8$ produce nine disjoint alternating cycles *by pure algebra*, giving immediately

$$\boxed{P(236) \ge 2^9 = 512}$$

without any matching enumeration. The remaining work for the exact count is only to exclude further cycles.

Lemma 7 (Alternating digraph at 236)**.** *Let D_p be the digraph with an arrow $i \to j$ whenever $i \ne j$ and $i + p(j)$ is a cube. Its strongly connected components comprise* 191 *singletons and exactly nine five-vertex sets*

$$S_t = \{10 + t, 46 + t, 71 + t, 137 + t, 198 + t\}, \qquad 0 \le t \le 8.$$

Within each S_t the nontrivial arrows form one simple directed 5*-cycle, without chords.*

Proof. This is a finite integer check using the only six possible cubes $8, 27, 64, 125, 216, 343$. The interval formula specifies every edge; the independent `count_236.py` verifier checks all 236 rows, the component partition, and the five-cycle properties with integer arithmetic. It invokes neither a matching solver nor a permanent-computation routine. □

Theorem 3 (Exact count and involution obstruction)**.**

$$\boxed{P(236) = 2^9 = 512.}$$

There is no cube-sum involution on $[1, 236]$.

Proof. Given any other cube-sum permutation q, $f = p^{-1} \circ q$ is a permutation whose arrows lie in D_p or are identity loops. Every cycle of f is confined to one strongly connected component. Each singleton is therefore fixed, while each S_t has exactly two choices: five fixed points or the unique directed 5-cycle. The choices are independent, so there are 2^9 permutations q.

For the central set $S_4 = \{14, 50, 75, 141, 202\}$, the reference permutation restricts to

$$14 \mapsto 50 \mapsto 75 \mapsto 141 \mapsto 202 \mapsto 14.$$

The only two possible restrictions of q to S_4 are this 5-cycle and its inverse, neither of which is an involution. Thus no q is an involution. □

The failure of involutivity at 236 is therefore a specific five-cycle obstruction, not a failure of cube-sum permutation existence. The proof uses only a fifteen-row interval template and an easily audited finite graph-decomposition lemma.

6 A computation-free exponential construction

A particularly simple weighted identity of four cubes produces more permutations than all the previous asymptotic constructions, with *no* finite matching certificate or eventual-existence theorem.

Proposition 2 (Weighted-cube odd-cycle principle)**.** *Let $q \geq 1$, and let A, B, X, Y be positive cubes such that*

$$(q+1)X + qY = A + 2qB.$$

For $0 \leq j \leq 2q$ put

$$D_j = \lceil j/2 \rceil (B - X) + \lfloor j/2 \rfloor (B - Y).$$

Assume $A \leq D_j \leq B - 2A$ for $j \geq 1$ and $|D_j - D_k| \geq A$ whenever $j \neq k$. Then for every integer $u \geq 1$,

$$\boxed{P((B-A)u^3) \geq 2^{Au^3-1}.}$$

In fact the family encodes $Au^3 - 1$ independent binary choices, each of which changes precisely $2q+1$ permutation positions.

Proof. Put $T = u^3$ and define the two-interval reflection $p(i) = AT - i$ for $i < AT$ and $p(i) = BT - i$ for $AT \leq i \leq (B-A)T$. The interval hypotheses make this a valid permutation with cube sums AT and BT. For each $1 \leq s < AT$, let $v_j = s + D_j T$ for $0 \leq j \leq 2q$. The gap conditions place v_0 in the first reflection block and the other vertices in the second, while ensuring that the translated cycles for different s are disjoint. The edge sums $v_j + p(v_{j+1})$ are alternately XT and YT for $j < 2q$. The final edge is a cube as well, since

$$v_{2q} + p(v_0) = (A + 2qB - qX - qY)T = XT.$$

Thus the $AT - 1$ cycles can be switched independently, proving the count and Hamming-distance statement. □

For $q = 1$ this recovers the three-cycle identity $2X + Y = A + 2B$. For $q = 2$ it becomes $3X + 2Y = A + 4B$. In particular,

$$3 \cdot 5^3 + 2 \cdot 8^3 = 3^3 + 4 \cdot 7^3$$

gives $(A, B, X, Y) = (27, 343, 125, 512)$ and offsets $(D_0, \ldots, D_4) = (0, 218, 49, 267, 98)$, satisfying every hypothesis. Consequently

$$\boxed{P(316u^3) \geq 2^{27u^3-1}}$$

for every positive integer u. This unifies the three- and five-cycle constructions through one algebraic identity.

Lemma 8 (Three-cycle identity gadget)**.** *Let A, B, X, Y be positive perfect cubes satisfying*

$$A + 2B = 2X + Y, \qquad A + \tfrac{1}{2}B \leq X \leq B - A.$$

For every positive integer u, put $T = u^3$ and $N = (B-A)T$. There are at least

$$\boxed{2^{AT-1}}$$

different permutations p of $[1, N]$ for which $i + p(i)$ is a perfect cube.

Proof. The displayed inequalities give $B \geq 4A$. Define a reference permutation by two reflections:

$$p(i) = \begin{cases} AT - i, & 1 \leq i < AT, \\ BT - i, & AT \leq i \leq (B - A)T. \end{cases}$$

The first interval maps to itself under the first reflection, and the second also maps to itself under the second reflection. Every original sum is AT or BT, both perfect cubes.

Set $d_1 = B - X$ and $d_2 = X - A$. The hypotheses imply

$$d_1 \geq A, \qquad d_2 - d_1 \geq A, \qquad d_2 + A = X \leq B - A.$$

For every $s = 1, \ldots, AT - 1$, the triple

$$C_s = (s,\ s + d_1T,\ s + d_2T)$$

lies inside $[1, N]$, with the second and third vertices in the second reflection interval. The three translated ranges are pairwise disjoint by the displayed gap inequalities. Thus all $AT - 1$ triples are pairwise vertex-disjoint.

Along the directed three-cycle $s \to s + d_1T \to s + d_2T \to s$, the three alternative matching sums are

$$\begin{aligned} s + p(s + d_1T) &= XT, \\ (s + d_1T) + p(s + d_2T) &= (A + 2B - 2X)T = YT, \\ (s + d_2T) + p(s) &= XT. \end{aligned}$$

Because T and A, B, X, Y are cubes, every new sum is a cube. Switching any subset of the $AT - 1$ disjoint cycles yields 2^{AT-1} pairwise different legal permutations. □

Theorem 4 (Explicit high-entropy cube-sum permutations)**.** *For every integer $u \geq 1$,*

$$\boxed{P(681707u^3) \geq 2^{148877u^3 - 1}.} \tag{4}$$

Consequently,

$$\boxed{\limsup_{n \to \infty} \frac{\log P(n)}{n} \geq \frac{148877 \log 2}{681707} > 0.} \tag{5}$$

This theorem is entirely computation-free, apart from elementary multiplication of four displayed integers.

Proof. Apply Lemma 8 with

$$\begin{aligned} A &= 53^3 = 148877, \quad & B &= 94^3 = 830584, \\ X &= 87^3 = 658503, \quad & Y &= 79^3 = 493039. \end{aligned}$$

Indeed,

$$A + 2B = 1810045 = 2X + Y,$$

and

$$A + \tfrac{1}{2}B = 564169 \leq X = 658503 \leq 681707 = B - A.$$

Hence the $AT - 1 = 148877u^3 - 1$ explicit cycles are disjoint. More precisely, each has the form

$$s \longrightarrow s + 172081u^3 \longrightarrow s + 509626u^3 \longrightarrow s,$$

with alternative sums $(87u)^3, (79u)^3, (87u)^3$. No matching search, graph decomposition, or auxiliary finite base case is involved. Taking logarithms along the sequence $n = 681707u^3$ proves the second claim. □

Corollary 1 (Linear-distance binary permutation codes)**.** *For $N = 681707u^3$ there is an injective map from $\{0,1\}^{148877u^3 - 1}$ into legal cube-sum permutations of $[1, N]$ that multiplies all Hamming distances by exactly three. Its distance enumerator is*

$$\left(1 + z^3\right)^{148877u^3 - 1}.$$

In particular, a greedy binary code subfamily has exponentially many legal permutations with pairwise Hamming distance proportional to N.

Proof. Each bit chooses whether to switch one of the pairwise disjoint three-cycles. A changed bit changes precisely three mapped positions. The distance identity and polynomial follow immediately. The greedy Hamming-ball argument from Corollary 3 applies unchanged. □

7 Power-block lifting and exponential growth on a subsequence

The nine disjoint alternating cycles at 236 can be amplified by a purely algebraic interval construction, yielding exponentially many solutions at infinitely many *different* lengths. The lifting mechanism works for every perfect-power exponent, not only cubes.

Lemma 9 (Independent-fiber perfect-power tensorization)**.** *Fix $d \geq 2$ and $M \geq 1$, and suppose $P_d(M) > 0$. For any odd positive integer t put $L = t^d$ and $A = (L-1)/2$. For each $r \in [1, L]$ choose independently a dth-power-sum permutation p_r of $[1, M]$. Define a permutation Q on the interval $I = [A+1, A+ML]$ by*

$$Q\big(A + (i-1)L + r\big) = A + (p_r(i) - 1)L + (L + 1 - r). \tag{6}$$

Then every $x + Q(x)$ is a perfect dth power, and the L independent choices give exactly

$$\boxed{P_d(M)^L}$$

different permutations of I.

If instead all fibers use the same p possessing h pairwise vertex-disjoint directed alternating cycles, then Q has at least hL pairwise vertex-disjoint directed alternating cycles.

Proof. The input positions have unique coordinates (i, r). For fixed r, the map $i \mapsto p_r(i)$ is a permutation of $[1, M]$, while the output residue $L + 1 - r$ identifies the original fiber r uniquely. Thus (6) is a bijection. For $x = A + (i-1)L + r$,

$$\begin{aligned} x + Q(x) &= 2A + L(i + p_r(i) - 1) + 1 \\ &= L(i + p_r(i)). \end{aligned}$$

If $i + p_r(i) = k_{i,r}^d$, then this sum equals $(tk_{i,r})^d$. The map from the tuple $(p_1, \ldots, p_L)$ to Q is injective, since all $p_r(i)$ are recovered from the output block indices; hence there are exactly $P_d(M)^L$ such interval permutations.

Finally, when every $p_r = p$, an alternating edge $i \to j$ of p gives $x_{i,r} + Q(x_{j,r}) = L(i + p(j))$ for each $x_{i,r} = A + (i-1)L + r$, again a perfect power. Every original alternating cycle consequently lifts to one cycle in each fiber, and disjoint fibers and original cycles remain disjoint. □

Theorem 5 (An exponential subsequence)**.** *For every odd integer $t \geq 9$, define*

$$\boxed{n_t = \frac{551t^3 - 1}{2}.}$$

Then

$$\boxed{P(n_t) \geq P\left(\frac{t^3 - 1}{2}\right) 512^{t^3} \geq 3 \cdot 2^{9t^3}.}$$

In particular,

$$\boxed{\limsup_{n \to \infty} \frac{\log P(n)}{n} \geq \frac{18 \log 2}{551} > 0.}$$

Proof. Take the explicit $M = 236$ permutation from Section 5, possessing $h = 9$ disjoint alternating 5-cycles. Let $L = t^3$, $A = (L-1)/2$, and $B = A + 236L = (473L - 1)/2$. Because $t \geq 9$, $A \geq 364 \geq 173$, so Theorem 1 supplies a valid permutation on the prefix $[1, A]$. By Theorem 2, $P(A) \geq 3$. The exact count $P(236) = 512$ from Theorem 3 and Lemma 9 provide 512^L distinct valid middle-block permutations on $[A+1, B]$, without needing to select a preferred alternating-cycle orientation.

Now let $C = (8t)^3 = 512L$ and set

$$n_t = C - B - 1 = \frac{551L - 1}{2}.$$

The final suffix $[B+1, n_t]$ has exactly $39L > 0$ elements. Define the permutation there by $x \mapsto C - x$. It permutes that suffix because $C = B + n_t + 1$, and every new sum is the cube C. Combining the three disjoint intervals produces a cube-sum permutation on $[1, n_t]$. The prefix and middle-block choices are independent, while the suffix is fixed. Consequently at least $P(A)\, 512^L$ distinct permutations are obtained, and $P(A) \geq 3$ completes the proof. The older $9L$ independent-cycle construction is a special subfamily of the new tensorized construction. □

Theorem 6 (Asymptotically optimal terminal reflection)**.** *Let $t \geq 9$ be odd and $L = t^3$. Let k_t be the least integer satisfying*

$$k_t^3 > 473L,$$

and define

$$\boxed{N_t = k_t^3 - \frac{473L+1}{2}.}$$

Then

$$\boxed{P(N_t) \geq P\left(\frac{L-1}{2}\right) 512^L \geq 3 \cdot 2^{9L}.}$$

Furthermore,

$$N_t = \left(\frac{473}{2} + o(1)\right) L \qquad (t \to \infty),$$

and therefore

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \geq \frac{18 \log 2}{473}.} \tag{7}$$

Proof. Take the 236-element tensorized gadget of Lemma 9. The valid prefix has length $A = (L-1)/2$ and the lifted middle block ends at $B = (473L-1)/2$. The strict inequality $k_t^3 > 473L = 2B+1$ guarantees the suffix interval $[B+1, N_t]$ is nonempty: indeed

$$N_t - B = k_t^3 - 473L > 0.$$

Reflection through the cube k_t^3 permutes that suffix since $B + N_t + 1 = k_t^3$. The middle blocks have 512^L independent choices, the prefix has $P(A)$ choices, and the suffix is fixed. The first bound follows, and $P(A) \geq 3$ because $A \geq 364$. By minimality,

$$(k_t - 1)^3 \leq 473t^3 < k_t^3,$$

so $k_t = 473^{1/3}t + O(1)$. Thus $k_t^3 = 473t^3 + O(t^2)$ and $N_t = (473/2)t^3 + O(t^2)$. Dividing $9L \log 2$ by N_t establishes (7). □

The fixed terminal cube $(8t)^3$ used in Theorem 5 is convenient for a simple closed formula $n_t = (551t^3 - 1)/2$, but is not asymptotically optimal. Selecting the *least* admissible terminal cube as above shrinks the suffix from $39t^3$ to $O(t^2)$, and improves the linear entropy coefficient from $18 \log 2/551$ to $18 \log 2/473$.

Corollary 2 (Multiplicity forces exponential subsequences)**.** *Fix $d \geq 2$. Suppose $P_d(n) > 0$ for all sufficiently large n and that $P_d(M) \geq 2$ for some $M \geq 1$. Choose any integer b with $b^d > 2M+1$. For all sufficiently large odd t, define*

$$n_t = \left(b^d - M - \tfrac{1}{2}\right)t^d - \tfrac{1}{2}.$$

Then

$$P_d(n_t) \geq P_d\left(\frac{t^d-1}{2}\right) P_d(M)^{t^d},$$

and in particular

$$\boxed{\limsup_{n\to\infty} \frac{\log P_d(n)}{n} \geq \frac{\log P_d(M)}{b^d - M - \frac{1}{2}} > 0.}$$

Proof. Let $L = t^d$, $A = (L-1)/2$ and $B = A + ML$. Eventual existence supplies a valid prefix on $[1, A]$ for all sufficiently large odd t, with $P_d(A)$ possible choices. Lemma 9 gives $P_d(M)^L$ independent middle-block choices. Now put $C = (bt)^d = b^d L$ and $n_t = C - B - 1$. Since $b^d > 2M+1$, the suffix $[B+1, n_t]$ is nonempty; reflection $x \mapsto C - x$ permutes it and makes every sum equal to C. The three intervals are invariant under the constructed permutation, and their choices are independent. Taking logarithms and dividing by n_t gives the bound. □

Theorem 7 (An exact binary cube-sum permutation code)**.** *For each odd $t \geq 9$, put $L = t^3$ and $n_t = (551L-1)/2$. There is an injective map*

$$\Phi_t : \{0,1\}^{9L} \longrightarrow \{\textit{cube-sum permutations of } [1, n_t]\}$$

such that for all bit strings u, v,

$$\boxed{d_{\mathrm{H}}(\Phi_t(u), \Phi_t(v)) = 5\, d_{\mathrm{H}}(u, v).} \tag{8}$$

In particular, relative to $\Phi_t(0)$, the exact Hamming-distance enumerator of this explicit family is

$$\boxed{\sum_{u \in \{0,1\}^{9L}} z^{d_{\mathrm{H}}(\Phi_t(u), \Phi_t(0))} = (1 + z^5)^{9L}.}$$

Proof. The proof of Theorem 3 identifies every 236-element cube-sum permutation with a unique nine-bit vector recording which of the nine disjoint directed five-cycles have been switched. Two different choices on one five-cycle change all five mapped values, while all untouched components agree. Thus their permutation Hamming distance is exactly five times the bitwise Hamming distance.

Take the L independent base-permutation fibers of Lemma 9, and concatenate their nine-bit selectors into $u \in \{0,1\}^{9L}$. Different fibers occupy disjoint input positions, so distances add exactly. Finally append a fixed valid prefix and the fixed cube-reflection suffix of Theorem 5. Neither changes the Hamming distances. The generating polynomial follows because each of the $9L$ bits independently contributes either distance zero or distance five. □

Corollary 3 (Positive-rate, linear-distance families)**.** *Let $H_2(x) = -x \log_2 x - (1-x)\log_2(1-x)$ be binary entropy. For every odd $t \geq 9$, there is a subfamily $\mathcal{F}_t$ of cube-sum permutations on $[1, n_t]$ such that*

$$|\mathcal{F}_t| \geq 2^{\,9L(1-H_2(1/4))}, \qquad d_{\min}(\mathcal{F}_t) > \frac{45L}{4}.$$

Consequently both $\log |\mathcal{F}_t| = \Omega(n_t)$ and $d_{\min}(\mathcal{F}_t) = \Omega(n_t)$.

Proof. Greedily select a binary word of length $N = 9L$, and delete all words within Hamming distance $\lfloor N/4 \rfloor$ of the selected word. A binary Hamming ball of this radius has at most $2^{NH_2(1/4)}$ members, so the greedy packing yields at least $2^{N(1-H_2(1/4))}$ codewords, with pairwise binary distance exceeding $N/4$. Applying Φ_t multiplies all distances by exactly five, while preserving cardinality. Since $n_t = (551L-1)/2$, both claimed linear asymptotics follow. □

Corollary 4 (Optimal general terminal choice)**.** *Under the assumptions of Corollary 2, the stronger universal estimate holds:*

$$\boxed{\limsup_{n \to \infty} \frac{\log P_d(n)}{n} \geq \frac{\log P_d(M)}{M + \frac{1}{2}}.}$$

Proof. Put $L = t^d$ with odd t and $B = ((2M+1)L - 1)/2$. Take the least integer k for which $k^d > (2M+1)L$. Since $k = (2M+1)^{1/d}t + O(1)$, the reflected suffix gives a terminal size $n = k^d - B - 1$ with $n = (M + \frac{1}{2})L + O(t^{d-1})$. The independent-fiber middle and eventually valid prefix yield $P_d(n) \geq P_d(M)^L$. Taking logarithms proves the bound. □

Theorem 8 (Power-divisor tensor recurrence)**.** *Fix an exponent $d \geq 2$ and a nonnegative integer n. For any odd positive integer t such that $L = t^d$ divides $2n+1$, set*

$$A = \frac{L-1}{2}, \qquad q = \frac{2n+1}{L}, \qquad M_b = b^d - \frac{q+1}{2}.$$

With $P_d(0) = 1$, we have

$$\boxed{P_d(n) \geq P_d(A) \sum_{\substack{b \geq 2,\ M_b \geq 0 \\ b^d < q}} P_d(M_b)^{\,L}.} \tag{9}$$

For $t = 1$, this is exactly the earlier one-step interval-reflection counting recurrence (now also allowing the empty prefix).

Proof. For each summand choose one permutation on $[1, A]$ and, independently, L permutations of $[1, M_b]$. Apply Lemma 9 on $[A+1, B]$, where $B = A + M_b L$. The equation defining M_b gives

$$B + n + 1 = b^d L = (bt)^d, \qquad n - B = (q - b^d)L > 0.$$

Thus the final interval $[B+1, n]$ is permuted by the reflection $x \mapsto (bt)^d - x$. All three pieces are valid and disjoint, producing $P_d(A)P_d(M_b)^L$ distinct permutations. Different b have different last entries $p(n) = (bt)^d - n$, so these families are disjoint and can be added. When $t = 1$, $L = 1$, $A = 0$ and $M_b = b^d - n - 1$, recovering the reflection recurrence. □

This formula gives a precise arithmetic description of the advantage and limitation of the tensor construction: a nontrivial block scale $t > 1$ requires the cubic divisibility condition $t^d \mid (2n+1)$. It does not, by itself, prove exponential growth at *every* length.

Proposition 3 (Arithmetic sparsity of nontrivial block lifting)**.** *The set of positive integers n for which some odd integer $t \geq 3$ satisfies $t^3 \mid (2n+1)$ has natural density*

$$\boxed{1 - \frac{8}{7\zeta(3)} = 0.0492487\ldots.}$$

Thus more than 95% of lengths have no nontrivial integral cube scale for the particular centered tensor construction used in Theorem 8.

Proof. An odd integer is cube-free precisely when it is divisible by no prime cube p^3, with p odd. By Möbius inclusion–exclusion, the number of positive $n \leq X$ for which $2n+1$ is cube-free equals

$$\begin{aligned}
\sum_{\substack{n \leq X \\ 2n+1 \text{ cube-free}}} 1 &= \sum_{\substack{d \leq (2X+1)^{1/3} \\ d \text{ odd}}} \mu(d) \#\{n \leq X : d^3 \mid 2n+1\} \\
&= X \sum_{\substack{d \geq 1 \\ d \text{ odd}}} \frac{\mu(d)}{d^3} + O(X^{1/3}) \\
&= X \prod_{p \text{ odd}} (1 - p^{-3}) + O(X^{1/3}) \\
&= \frac{8X}{7\zeta(3)} + O(X^{1/3}).
\end{aligned}$$

Every odd composite $t \geq 3$ has an odd prime divisor, so $t^3 \mid 2n+1$ is possible for some $t \geq 3$ exactly when $2n+1$ is not cube-free. Taking complements proves the claim. □

This is a limitation of the *specific centered tensor construction*, not an upper bound on $P(n)$ or a no-go theorem for other embeddings. In particular, the exponential-in-n subsequence result cannot by itself be promoted to a uniform exponential lower bound using the same integral scales for every n.

Lemma 10 (A denser finite alternating-cycle gadget)**.** *There exists a cube-sum permutation of $[1, 362]$ admitting at least 40 pairwise vertex-disjoint alternating cycles. Consequently,*

$$\boxed{P(362) \geq 2^{40}.}$$

Proof. An explicit certificate in `certificates/forty_cycles_362.json` describes a valid permutation using 43 integer interval-reflection rows $p(i) = c - i$ and lists 40 alternating directed cycles on 182 distinct vertices. Each listed edge is checked by the single integer identity $i + p(j) = k^3$. These cycles are vertex-disjoint and can be switched independently. The standalone `verify_362.py` checker directly verifies every row and edge without any matching solver or permanent algorithm. □

Theorem 9 (Improved explicit exponential subsequence)**.** *For every odd integer $t \geq 7$, let*

$$N_t = \frac{733t^3 - 1}{2}.$$

Then

$$\boxed{P(N_t) \geq P\left(\frac{t^3-1}{2}\right) 2^{40t^3} \geq 2^{40t^3}.}$$

In particular,

$$\boxed{\limsup_{n\to\infty} \frac{\log P(n)}{n} \geq \frac{80\log 2}{733} > 0.0756.}$$

Proof. In Theorem 8 take $d = 3$, $M_b = 362$, $b = 9$. Then $b^3 = 729$ and $q = 2\cdot 729 - 2\cdot 362 - 1 = 733$, so $b^3 < q$ and $n = N_t$. At $t = 7$ the prefix length $A = (7^3-1)/2 = 171$ is a known positive index from the finite classification. For every larger odd t, $A \geq 364$, so the existence theorem applies. Hence $P(A) > 0$ and

$$P(N_t) \geq P(A)P(362)^{t^3} \geq 2^{40t^3}.$$

Since $N_t \sim (733/2)t^3$, division by N_t gives the stated exponential coefficient. □

The constant here is stronger than the earlier 236-seed coefficient $18\log 2/551$. It is a rigorously certified *lower bound*, not a claim of optimal asymptotic entropy.

This gives an *exponential limsup*, not an exponential lower bound for all n. In particular, it is logically distinct from the uniform, near-critical reflection estimates in Section 9.

8 Many permutations: an exponential cube-root bound

Besides qualitative multiplicity, the reflection family yields a strong quantitative statement. Let $K(n) = \lfloor \sqrt[3]{2n} \rfloor$, with the floor defined using integer comparisons (not floating point).

Lemma 11 (Three disjoint embeddings)**.** *If $K(n) = k \geq 27$, each of $k^3, (k-1)^3, (k-2)^3$ gives a reflection extension from a predecessor $m \geq 173$ with $m < n$, and all three predecessors obey $K(m) \geq k - 6$.*

Proof. By definition, $k^3 \leq 2n < (k+1)^3$. For the smallest of these cubes, set $m = (k-2)^3 - n - 1$. Then

$$2m > 2(k-2)^3 - (k+1)^3 - 2 = k^3 - 15k^2 + 21k - 19.$$

The right side exceeds $(k-6)^3$ for $k \geq 27$, since the difference is

$$3k^2 - 87k + 197 > 0 \quad (k \geq 27).$$

Thus $2m > (k-6)^3 \geq 21^3$, which implies $m \geq 173$ and $K(m) \geq k - 6$. Also $(k-2)^3 < k^3 \leq 2n$, so $m < n$. The other two cubes give larger predecessors and satisfy the same bounds. Distinct cubes yield disjoint matching families by Lemma 1. □

Theorem 10 (A096904 quantitative bound)**.** *For all integers $n \geq 173$, with $q(n) = \max\{0, \lfloor (K(n) - 27)/6 \rfloor\}$,*

$$\boxed{P(n) \geq 3^{q(n)}}.$$

In particular

$$\log P(n) \;\geq\; \frac{\log 3}{6}(2n)^{1/3} - O(1),$$

so $P(n)$ grows at least exponentially in $n^{1/3}$.

Proof. If $K(n) < 27$ then $q(n) = 0$, and Theorem 1 supplies $P(n) \geq 1$. For $K(n) \geq 27$, Lemma 11 provides three distinct extensions from smaller predecessors $m_i \geq 173$ with $K(m_i) \geq K(n) - 6$. Thus $q(m_i) \geq q(n) - 1$. By strong induction and the disjointness of the three extension images,

$$P(n) \geq \sum_{i=1}^{3} P(m_i) \geq 3 \cdot 3^{\max(0, q(n)-1)} \geq 3^{q(n)}.$$

The displayed logarithmic asymptotic follows immediately. □

9 A universal reflection-entropy bound

The fixed three-branch construction above is only one instance of a general branching mechanism. In fact, one may use *any fixed number* of the largest available cubes and obtain a sharp limiting coefficient for this family of lower bounds.

For each integer $s \geq 2$, let $\rho_s \in (1,2)$ be the unique solution of

$$\sum_{h=1}^{s} \rho_s^{-h} = 1, \qquad B_s = 2s^2 + 4s - 1. \tag{10}$$

Let $K(n) = \lfloor \sqrt[3]{2n} \rfloor$ as before.

Theorem 11 (Reflection entropy)**.** *For all $s \geq 2$ and $n \geq 173$,*

$$\boxed{P(n) \geq \rho_s^{(K(n)-B_s)/2}.} \tag{11}$$

In particular,

$$\boxed{\liminf_{n\to\infty} \frac{\log P(n)}{(2n)^{1/3}} \;\geq\; \frac{\log 2}{2}.} \tag{12}$$

Equivalently, for every $\varepsilon > 0$ there is a constant C_ε such that

$$P(n) \geq \exp\Big(\big(\tfrac{1}{2}\log 2 - \varepsilon\big)\,(2n)^{1/3} - C_\varepsilon\Big) \quad (n \geq 173).$$

Proof. Fix s and write $B = B_s$, $\rho = \rho_s$. We prove (11) by strong induction. For $K(n) < B$, the right side is at most 1, and $P(n) \geq 1$ by Theorem 1.

Now suppose $k = K(n) \geq B$. For each $j = 0, \ldots, s-1$, take the cube $c_j = (k-j)^3$ and put $m_j = c_j - n - 1$. Since $k^3 \leq 2n < (k+1)^3$, the following exact polynomial inequality holds:

$$\begin{aligned} 2m_j - (k-2j-2)^3 &\geq 2(k-j)^3 - (k+1)^3 - 1 - (k-2j-2)^3 \\ &= 3k^2 - (6j^2+24j+15)k \\ &\quad + 6j^3 + 24j^2 + 24j + 6 \\ &\geq 0. \end{aligned}$$

Dividing the final polynomial by 3, its coefficient of $-k$ is $2j^2+8j+5$ and its constant term is positive. Hence the last inequality follows from $k \geq B \geq 2j^2+8j+5$. Also $k - 2j - 2 \geq k - 2s \geq 8$, so $m_j \geq 256 \geq 173$ and $K(m_j) \geq k - 2(j+1)$. Each $c_j \leq k^3 \leq 2n$, so $m_j < n$.

The s reflection extensions are pairwise disjoint by their distinct last images, as in Lemma 1. Using the strong induction hypothesis, we obtain

$$\begin{aligned} P(n) &\geq \sum_{j=0}^{s-1} P(m_j) \\ &\geq \sum_{j=0}^{s-1} \rho^{(K(m_j)-B)/2} \\ &\geq \rho^{(k-B)/2} \sum_{j=0}^{s-1} \rho^{-(j+1)} = \rho^{(k-B)/2}. \end{aligned}$$

This proves the first assertion. The ρ_s are strictly increasing and converge to 2, since their defining finite geometric sums approach $\sum_{h\geq 1} 2^{-h} = 1$. For each fixed s, B_s is a constant independent of n and $K(n) = (2n)^{1/3} + O(1)$. Therefore the ratio on the left side of (12) is at least $\frac{1}{2}\log\rho_s$ in the limit. Letting $s \to \infty$ proves (12). □

Corollary 5 (Other power sums, conditional on existence)**.** *For an integer $d \geq 2$, let $P_d(n)$ count permutations of $[1,n]$ in which every $i + p_i$ is a perfect dth power. If $P_d(n) > 0$ for all sufficiently large n, then*

$$\liminf_{n\to\infty} \frac{\log P_d(n)}{(2n)^{1/d}} \geq \frac{\log 2}{2}.$$

Proof. For any fixed s and $j = 0, \dots, s-1$, write $k = \lfloor (2n)^{1/d} \rfloor$ and use $c_j = (k-j)^d$, $m_j = c_j - n - 1$. The binomial theorem gives, uniformly over these finitely many j,

$$2(k-j)^d - (k+1)^d - (k-2j-2)^d = dk^{d-1} + O_{d,s}(k^{d-2}) > 1$$

for all sufficiently large k. It follows that m_j is a smaller positive length in the eventual-existence range and $\lfloor (2m_j)^{1/d} \rfloor \geq k - 2j - 2$. The distinct reflection families obey the same s-branch induction as Theorem 11, with a suitable constant initial offset in k. The ρ_s tend to 2 as before. □

Theorem 12 (A near-critical lower bound with explicit loss)**.** *For $K(n) = \lfloor \sqrt[3]{2n} \rfloor$ and every $n \geq 173$,*

$$\boxed{P(n) \geq \frac{127}{K(n)}\, 2^{(K(n)-127)/2}.} \tag{13}$$

More generally, for every real $\varepsilon > 0$ there exists $C_\varepsilon > 0$ such that

$$\boxed{P(n) \geq C_\varepsilon \frac{2^{K(n)/2}}{K(n)^\varepsilon} \qquad (n \geq 173).} \tag{14}$$

Proof. Write $k = K(n)$. The exact cubic polynomial inequality in the proof of Theorem 11 works whenever

$$k \geq B_s = 2s^2 + 4s - 1, \qquad s \geq 2;$$

it supplies s *disjoint* reflection embeddings with predecessor indices $m_h \in [173, n-1]$ and $K(m_h) \geq k - 2h$ for $h = 1, \dots, s$.

First set $F(k) = 2^{k/2}/k$, which is increasing for all integer $k \geq 3$. For $k \geq 128$ choose $s = \lfloor \log_2 k \rfloor$. Then $s \geq 7$ and $k \geq 2^s \geq B_s$: the last inequality holds at $s = 7$, since $128 \geq 125$, and is preserved upon doubling, because $2B_s \geq B_{s+1}$ for all $s \geq 7$. Moreover $k - 2s \geq 8$, so every predecessor is in the increasing range of F. By the reflection counting lemma,

$$\begin{aligned} \frac{\sum_{h=1}^s F(k-2h)}{F(k)} &= \sum_{h=1}^s \frac{2^{-h}k}{k-2h} \\ &\geq \sum_{h=1}^s 2^{-h}\left(1 + \frac{2h}{k}\right) \\ &\geq 1 - 2^{-s} + \frac{2}{k} > 1, \end{aligned}$$

using $\sum_{h=1}^s h2^{-h} \geq 1$ for $s \geq 2$ and $k < 2^{s+1}$. For $k < 128$, Theorem 1 ensures $P(n) \geq 1$. Because F is increasing for $k \geq 7$, this implies $P(n) \geq F(k)/F(127)$. Strong induction using the above inequality extends the same bound to all $k \geq 128$. Since $F(127) = 2^{127/2}/127$, this is exactly (13).

For the more general result, fix $\varepsilon > 0$ and define $F_\varepsilon(k) = 2^{k/2}/k^\varepsilon$. It is increasing for all sufficiently large k. For sufficiently large k, set $s = \lceil \log_2(k/\varepsilon) \rceil$, so $s = O_\varepsilon(\log k)$, $k \geq B_s$ and $2^{-s} \leq \varepsilon/k$. Each predecessor is in the increasing range of F_ε. By the elementary Bernoulli inequality,

$$\left(\frac{k}{k-2h}\right)^\varepsilon \geq 1 + \frac{2\varepsilon h}{k}.$$

Consequently

$$\begin{aligned} \frac{\sum_{h=1}^s F_\varepsilon(k-2h)}{F_\varepsilon(k)} &\geq 1 - 2^{-s} + \frac{2\varepsilon}{k}\sum_{h=1}^s h2^{-h} \\ &\geq 1 - 2^{-s} + \frac{2\varepsilon}{k} > 1. \end{aligned}$$

Choose an initial K_ε large enough for all these inequalities to hold at $k \geq K_\varepsilon$. By positivity, the bound $P(n) \geq C_\varepsilon F_\varepsilon(k)$ holds for $k < K_\varepsilon$ on taking $C_\varepsilon = 2^{-K_\varepsilon/2}$. The same strong induction proves it for every larger k, yielding (14). □

These bounds approach the *critical* reflection coefficient $\frac{1}{2}\log 2$ with a logarithmic (indeed arbitrarily small polynomial) loss. Their proofs use no new computer-assisted certificate beyond the existence theorem.

The coefficient $(\log 2)/2$ is a *proved lower bound*, not an asserted exact exponential growth rate. The argument needs only the qualitative existence Theorem 1; it makes no further use of numerical matching data.

10 An efficient constructive algorithm

The proof is also algorithmic. Given $n > 364$, choose the *least* cube $c = k^3 \geq n + 174$ and put $m = c - n - 1$. The cube-gap lemma ensures $173 \leq m < n$. Minimality gives

$$(k-1)^3 < n + 174 \leq k^3 \quad \Longrightarrow \quad 173 \leq m \leq 3k^2 - 3k + 173 = O(n^{2/3}).$$

Repeat until the remaining size is in the certified interval $173 \leq m \leq 364$, then fill each recorded interval by the explicit reflection $p_i = c - i$.

Proposition 4 (Linear-output construction)**.** *A valid cube-sum permutation of any $n \geq 173$ can be constructed in $O(n)$ unit-cost arithmetic operations, with $O(\log \log n)$ reflection layers.*

Proof. The predecessor bound $m = O(n^{2/3})$ implies $\log m \leq (2/3) \log n + O(1)$, so the number of iterations before the fixed-size base is $O(\log \log n)$. The returned intervals are disjoint and their union, together with the base interval, is $\{1, \ldots, n\}$. Therefore exactly n entries are written. Each exact integer cube root can be computed using $O(\log n)$ comparisons, which contributes $O(\log n \log \log n) = o(n)$ additional unit-cost operations. □

The accompanying `construct_linear.py` implements this procedure with exact integer cube roots, and can also display the recursion plan without allocating the full output.

11 Reproduction and limitations

All programs require only Python 3 standard-library modules:

```
python papers/A006063/verify.py
python papers/A006063/independent_match.py
python papers/A006063/verify_extended.py
python papers/A006063/construct_linear.py
python papers/A006063/verify_word_paths.py
python papers/A006063/count_236.py
```

The archived data consist of 109 core witnesses, the legacy 192-row all-base cross-check, 48 small positive permutations, 124 negative Hall sets, and compact pairs of alternating-cycle certificates. The infinite inductive and counting lemmas are mathematical arguments rather than extrapolations from numerical experiments.

The status is a *computer-assisted proof candidate* pending independent scrutiny of the finite certificate, its interpretation, and literature priority. The present manuscript does not compute the exact values of $P(n)$ in general, nor does it claim external peer review.

A All 191 short alternating-path words

The integer at the left is the new size n; the following one-digit symbols are the successive cube roots r_j in Lemma 4. Together with the explicit seed in Lemma 3, these words form an independently and locally checkable existence certificate for $173 \le n \le 364$.

174: 635555363635555363633555536
175: 6336
176: 65555556
177: 6364636
178: 6364625555556
179: 74635555555547
180: 74474247447
181: 6554746474556
182: 655556
183: 663647
184: 66
185: 66264746366
186: 66266
187: 7447
188: 7455665547
189: 666363647447447
190: 647
191: 6665547556
192: 75257
193: 7575555757
194: 66655666
195: 6264757
196: 663666626
197: 75746264757
198: 65575264757
199: 666
200: 66633666
201: 64747574757
202: 6474752556
203: 663466666
204: 757424757
205: 65755756
206: 6666446666
207: 666665546666
208: 64757
209: 6443756
210: 757464757
211: 6666357
212: 66664666
213: 757475757
214: 653756
215: 6666556
216: 7555666666665557
217: 75757375375757375757
218: 7555757573573757575557
219: 757455757464757
220: 75757554665575757
221: 757573753666375757
222: 7375757
223: 757455755557554757
224: 7363753637
225: 7557357447557
226: 75447557
227: 757537525735757
228: 7524665566666657
229: 746664667
230: 752757573637557
231: 75666637575725257
232: 757536357556666665757
233: 75666666557537
234: 7375757256666667
235: 77
236: 7466466466466466466475727574 7
237: 7375747
238: 74664747
239: 73444667
240: 7466466444667
241: 7337
242: 734664667
243: 73527537534667
244: 73667
245: 746666667
246: 766375375375266667
247: 7666666666666666667
248: 76666266667
249: 766662757475757266667
250: 747
251: 766665736666667
252: 747573375747
253: 7474466375747
254: 74744664664747
255: 74757377
256: 8
257: 76666667
258: 747525747
259: 85666658566277
260: 85662666666667
261: 7475375758
262: 8558
263: 7736666444758
264: 7737574744758
265: 8558534758
266: 766664758
267: 85585757377
268: 7477
269: 766666677477
270: 8575247477
271: 8575266667
272: 7475375375758575758
273: 855857573757447
274: 85266258
275: 7437447
276: 74586666666658
277: 868
278: 7447
279: 868525377
280: 773368
281: 76666477
282: 72755857477
283: 727
284: 777558
285: 727575868
286: 85475758
287: 76658
288: 7577666658
289: 8666857537
290: 8558
291: 856666667
292: 86468
293: 8558667
294: 737575727737
295: 86477737
296: 7775486477
297: 83747775868
298: 7577738
299: 7775468
300: 8646368585486868
301: 7368666468
302: 76644867
303: 734775557
304: 736868
305: 8677777
306: 7586666856768
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359: 8468
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361: 8586855588
362: 883477778
363: 8528686488
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